

able, long-term customer relationships, and is one that is slowly catching up. Again, utilising the customer's social media presence and the vast amount of data that comes with it, has the potential to improve both effectiveness and efficiency of marketing efforts. This will not only reduce customer frustration because of unwanted promotional offers and events, but also help improve customer retention for banks and better customer relations.

Tracking transactions

Taking the usage of transaction data a step further, banks can have unique insight into how, where, with whom and when customers are spending money, particularly by pairing this information with the social media activity of the customer. By analysing such information, banks can build an insight into customer intelligence, their spending patterns and behaviour. Unfortunately, bigger banks have been slow on the uptake, and their current usage of the treasure trove of data that is available out in the open has been less than satisfactory. On the other hand, big e-commerce players such as Amazon have successfully harnessed the power of social media and use the information to provide highly personalised services, such as product recommendations.

Another major application, one that was thrust into the spotlight after 2008, is the compliance and regulatory aspect of banking, that can be handled by engaging with data science techniques to maintain records of transactions and document everything that goes into each swap trade by implementing a deal monitoring system. Moreover, regulatory authorities also have the option of harnessing big data to fish out offenders and control the working of a sector that spawns petabytes of data a day.

Trading and portfolio management

But the most widely used function utilising the power of big data, is that of brokers and traders dealing on the stock market. The sheer efficiency and speed of a big data architecture makes it the perfect candidate for the foundation of a system, which provides the future prices of shares, securities and commodities and also predicts future market conditions. In fact, many automated systems that are currently being utilised, are capable of analysing the available data to buy and sell various securities on the market. This particular application now has its own field of study called “quantitative trading”, and has taken world economies by storm. In today's context, when trading takes place by the second rather than minute, these systems